

Public Debt and the Price of Fiscal Uncertainty: Attention and Inflation Volatility

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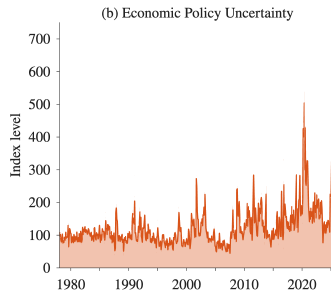
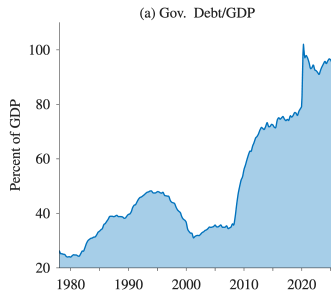
CESifo Area Conference on Macro with Micro Data

June 2026

¹The views expressed in this paper are those of the authors and do not necessarily represent the views or policies of the Board of Governors of the Federal Reserve System or its staff.

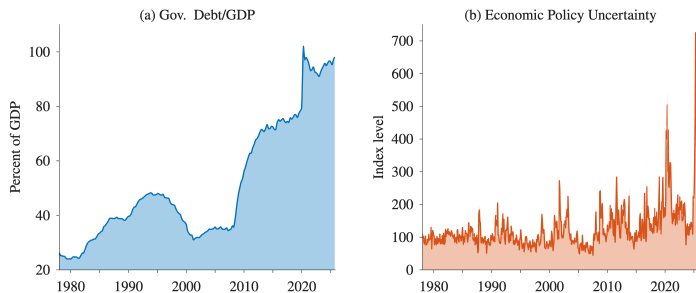
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- ▶ Public debt increased significantly in advanced economies in recent decades
- ▶ Uncertainty about policy and fiscal outlook has also increased



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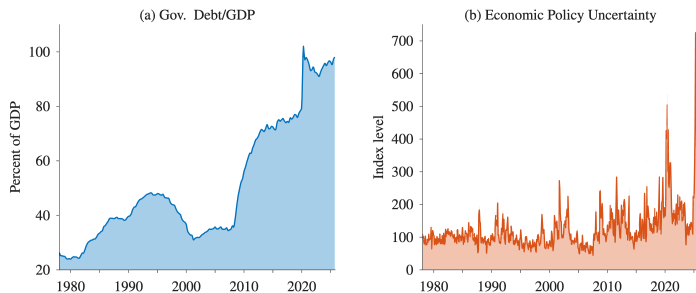
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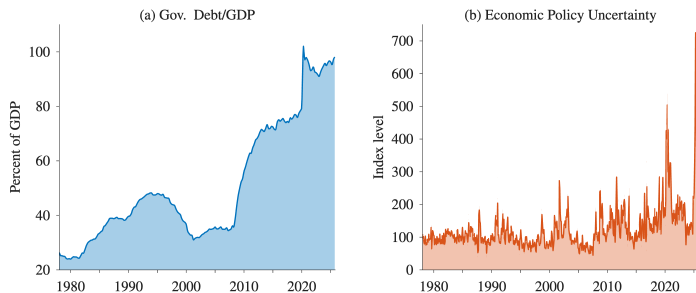


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Q: How does fiscal policy affect firms' price-setting behavior and inflation?

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 2. aggregate inflation

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 - ▶ [higher uncertainty amplifies fiscal shocks, especially at high debt levels](#)

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 - ... because it raises attention toward fiscal policy, especially at high debt levels
 - ▶ higher uncertainty amplifies fiscal shocks, especially at high debt levels $\hookrightarrow$ [endogenous attention renders the effects of fiscal policy state dependent](#)

Literature

- ▶ Rational inattention, incomplete information, price setting: Sims (2003,2010), Mackowiak/Wiederholt (2009), Mackowiak et al. (2023), Afrouzi (2024), Afrouzi/Yang (2021), Afrouzi et al. (2026), Yang (2022), Pfäuti (2025, 2026), Wu (2025), Reis (2006), Flynn/Sastry (2024), Song/Stern (2025), Stevens (2019),...
- ↪ **Contributions:** role of fiscal policy for attention and pricing decisions, solution method: dynamic rational inattention problem with free public signals
- ▶ Fiscal uncertainty, fiscal adjustment, inflation: Sargent/Wallace (1981), Leeper (1991), Sims (1994), Woodford (2001), Cochrane (1998), Bhattarai et al. (2014), Bianchi/Melosi (2017,2019), Eusepi/Preston (2018), Angeletos et al. (2025), Bayer et al. (2025),...
- ↪ **Contributions:** endogenous attention, role of price setters
- ▶ Estimated passthrough of costs to prices: Amiti et al. (2019): 0.6% (Belgium), Dogra et al. (2023): 0.6% (US), Gödl-Hanisch/Menkhoff (2025): $\approx 0.5\%$ after 6 months (Germany), Dedola et al. (2025): 0.3% (Germany, tax shocks)
- ↪ **Contributions:** passthrough (of expected cost) varies with debt/GDP in the Euro Area

Outline

1. Model
2. Analytical Results
3. Empirics
4. Quantitative Results

Model Overview

- ▶ Discrete time, $t = 0, 1, \dots$
- ▶ Households: consume, work, save in government bonds, receive profits ▶ [Details](#)
- ▶ Fiscal policy: issues *nominal* debt, raises revenue taxes to repay debt over time
- ▶ Monetary policy follows a standard Taylor rule
- ▶ Final good producer aggregates intermediate goods into final consumption good

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- ▶ Final good producer aggregates intermediate goods into final consumption good
- ▶ Intermediate firms:
 - ▶ Monopolistic competition, price setters
 - ▶ Pay revenue taxes ▶ Profit & Payroll Taxes
 - ▶ (Rationally) inattentive

Monetary and Fiscal Policy

Monetary authority follows a standard Taylor rule:

$$(1 + i_t) = (1 + i) (1 + \pi_t)^{\phi_\pi}$$

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$$B_t = (1 + i_{t-1})B_{t-1} - \tau_{r,t} \int_0^1 P_{j,t} Y_{j,t} dj, \quad t \geq 0$$

and implements the tax rule:

$$\tau_{r,t} = \tau_r \left(\frac{B_{t-1}}{P_{t-1}} \times \frac{P}{B} \right)^{\phi_b} e^{\nu_t} \quad (\text{or linearized: } \tau_{r,t} = \phi_b b_{t-1}^R + \nu_t)$$

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Steady state debt/GDP ratio: $\bar{i}\mathcal{B} = \mathcal{T}_r$

Price-Setting Firms

- ▶ Linear production function: $Y_{j,t} = Z_t L_{j,t}$, with productivity shocks $Z_t = e^{z_t}$, $z_t = \rho_z z_{t-1} + \epsilon_t^z$, $\epsilon_t^z \sim \mathcal{N}(0, \sigma_z^2)$
- ▶ Marginal cost: $MC_t \equiv \frac{W_t}{Z_t}$
- ▶ Under full information, firm j chooses $P_{j,t}$ to solve:

$$\max_{P_{j,t}} [(1 - \tau_{r,t})P_{j,t} - MC_t] \left(\frac{P_{j,t}}{P_t} \right)^{-\eta} Y_t$$

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... but firms are rationally inattentive

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- ▶ Firms observe history of aggregate prices for free
- ▶ LQG: problem boils down to choosing posterior uncertainty about $\vec{x}_t$

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How Fiscal Policy Shapes Firms' Attention

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- ▶ The (linearized) **desired price** absent information frictions is given by:

$$p_j^{\diamond} = \text{mc} + \gamma(\mathcal{B})\tau_r, \text{ with } \gamma(\mathcal{B}) = \frac{i\mathcal{B}}{1 - i\mathcal{B}}$$

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- ▶ $\gamma(\mathcal{B})$: sensitivity of the desired price to taxes, increasing and convex in $\mathcal{B}$
- ▶ Deviations from this desired price result in losses:

$$L(p_j; mc, \tau_r) = \frac{\mathcal{S}(\mathcal{B})}{2}(p_j - p_j^{\diamond})^2, \text{ where } \mathcal{S}(\mathcal{B}) = (1 - i\mathcal{B})(\eta - 1)$$

- ▶ $\mathcal{S}(\mathcal{B})$: curvature of profit function (stakes from mispricing), decreasing in $\mathcal{B}$

Solving Firms' Attention Problem

Focus for now on a static, partial equilibrium example. If we assume

$$\begin{pmatrix} mc \\ \tau_r \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{mc}^2 & 0 \\ 0 & \sigma_{\tau}^2 \end{pmatrix}\right)$$

we can write the firm's attention choice problem as:

$$\max_{\sigma_{post}^2 \leq \sigma_{p^\diamond}^2} -\frac{\mathcal{S}(\mathcal{B})}{2} \sigma_{post}^2 - \frac{\omega}{2} \ln \left(\frac{\sigma_{p^\diamond}^2}{\sigma_{post}^2} \right) \quad (1)$$

with

- ▶ $\sigma_{p^\diamond}^2 = \sigma_{mc}^2 + \gamma(\mathcal{B})^2 \sigma_{\tau}^2$ being the prior uncertainty and
- ▶ $\sigma_{post}^2 \equiv \mathbb{E}[(p_j^\diamond - \mathbb{E}[p_j^\diamond | S_j^0])^2]$ the posterior uncertainty about $p_j^\diamond$

Optimal Attention

Proposition

A firm's optimal posterior uncertainty and optimal price are given by

$$\sigma_{post}^2 = \min \left\{ \frac{\omega}{\mathcal{S}}, \sigma_{p^\diamond}^2 \right\}, \text{ and } p_j^* = \lambda(p_j^\diamond + \varepsilon_j)$$

where $\lambda = \max \left\{ 1 - \frac{\omega}{\mathcal{S}\sigma_{p^\diamond}^2}, 0 \right\}$ denotes the firm's attention.

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Attention is higher when

1. the information cost ω is lower
2. the stakes $\mathcal{S}$ are higher
3. or when the prior uncertainty $\sigma_{p^\diamond}^2$ is higher

How Fiscal Policy Shapes Firms' Attention

Theorem

When $\lambda > 0$, the effect of the debt/GDP ratio $\mathcal{B}$ on optimal attention λ is

$$\frac{\partial \lambda}{\partial \mathcal{B}} = \frac{\omega}{\mathcal{S}(\mathcal{B})\sigma_{p^\diamond}^2} \left[\underbrace{\frac{\partial \mathcal{S}}{\partial \mathcal{B}} \frac{1}{\mathcal{S}}}_{<0} + \underbrace{\frac{\partial \sigma_{p^\diamond}^2}{\partial \mathcal{B}} \frac{1}{\sigma_{p^\diamond}^2}}_{>0} \right]$$

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Furthermore, for $\mathcal{B} \in (0, \frac{1}{i})$, there exists a unique $\mathcal{B}^* = \frac{1}{i} \left(1 - \sqrt{\frac{1}{1 + \sigma_{mc}^2/\sigma_\tau^2}} \right)$ such that $\frac{\partial \lambda}{\partial \mathcal{B}}|_{\mathcal{B}=\mathcal{B}^*} = 0$ and for $\mathcal{B} < \mathcal{B}^*$, $\frac{\partial \lambda}{\partial \mathcal{B}} < 0$ and for $\mathcal{B} > \mathcal{B}^*$, $\frac{\partial \lambda}{\partial \mathcal{B}} > 0$.

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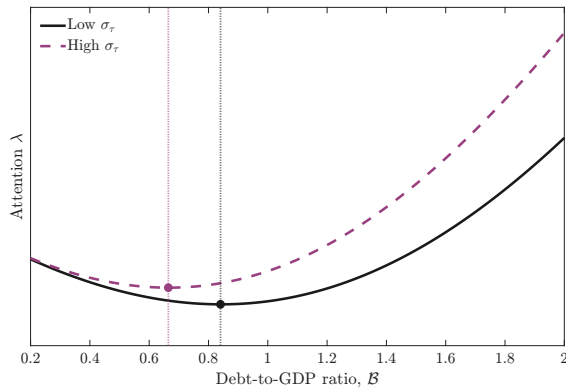
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↪ **Attention is U-shaped in $\mathcal{B}$**

↪ $\mathcal{B}^*$ is decreasing in $\frac{\sigma_\tau^2}{\sigma_{mc}^2}$

U-Shaped Attention: Numerical Example



Calibration: $\eta = 6, \bar{i} = 0.06, \omega = 4.5 \cdot 10^{-5}, \sigma_{mc} = 0.005$, Low $\sigma_{\tau} = 0.014$ and high $\sigma_{\tau} = 0.018$.

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Data

- ▶ ECB/EC *Survey on the Access to Finance of Enterprises* (SAFE):
 - ▶ 11 euro-area countries, 10 quarterly waves between 2023Q3 and 2026Q1
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Data

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Regression:

$$\Delta p_{i,c,t}^e = \beta_0 \Delta mc_{i,c,t}^e + \beta_1 \Delta mc_{i,c,t}^e \cdot \mathcal{B}_{c,t} + X'_{i,c,t} \Gamma + \alpha_{c,t} + \varepsilon_{i,c,t}.$$

- ▶ $\mathcal{B}_{c,t}$: country c 's debt/GDP ratio, lagged one quarter (source: Eurostat)
- ▶ X : size, sector, turnover bracket, expectations about wages and employment, and changes in turnover, profits, interest expenses, debt/assets, other costs
- ▶ $\alpha_{c,t}$: country $\times$ wave fixed effects; SEs clustered by country

What the theory predicts

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- ▶ **Rational inattention:** $\beta_{RI}(\mathcal{B}) = 1 + \frac{\gamma(\mathcal{B})^2 \sigma_{\tau}^2}{\sigma_{mc}^2}$

$\Rightarrow$ RI model predicts increasing passthrough in $\mathcal{B}$ ▶ DRS

Empirical results

	$\hat{\beta}_0$	$\hat{\beta}_1 \cdot 100$
Baseline	0.194*** (0.023)	0.0719*** (0.0298)
Implied passthrough:		
at $\mathcal{B} = 33\%$		0.218
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Robustness across 41 specs:

- ▶ $\hat{\beta}_1 > 0$ in **41/41** specs
- ▶ Significant at 10% in **39/41**
- ▶ Significant at 5% in **37/41**

Specs include: control combinations, leave-one-country-out, weighted/unweighted, percentile winsorization, HC1 SEs, median, Huber. ▶ [Details](#)

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- ▶ Robust to using [taxes on production & imports](#) instead of debt: implied passthrough rises from 0.21 at lowest tax-to-GDP ratio to 0.31 at highest ▶ [Details](#)

Outline

1. Model
2. Analytical Results
3. Empirics
4. Quantitative Results

Attention is U-shaped in $\mathcal{B}$

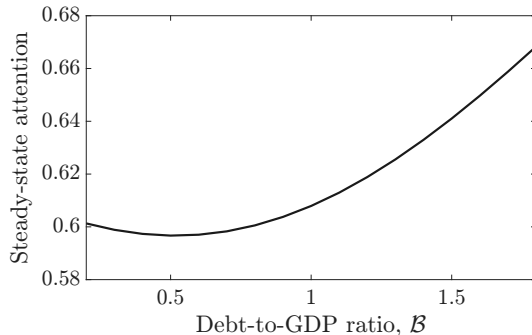
We now solve the dynamic rational inattention GE model with public signals

► Solution and Calibration Details

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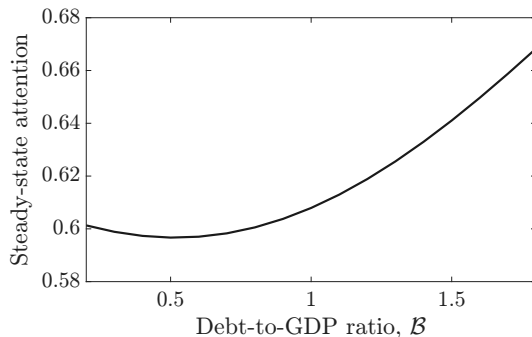


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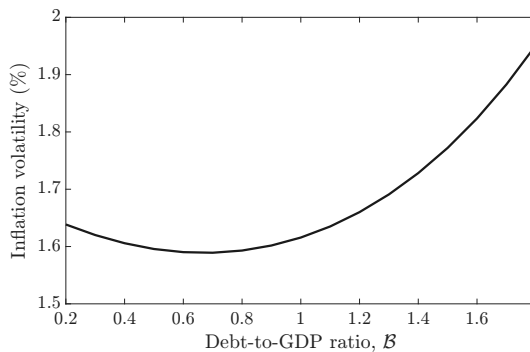
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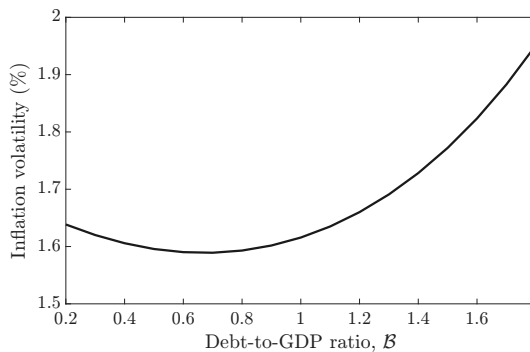


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- Attention is again U-shaped in $\mathcal{B}$ (the static/partial-equilibrium mechanism “survives”)
- What about inflation volatility?

Inflation volatility is also U-shaped

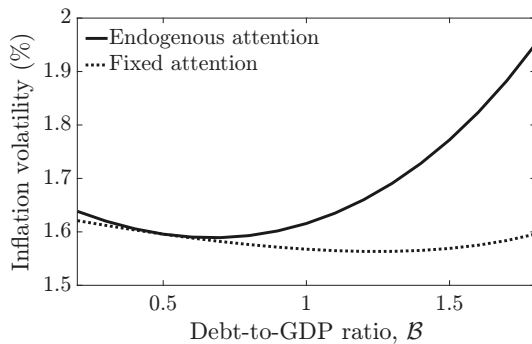


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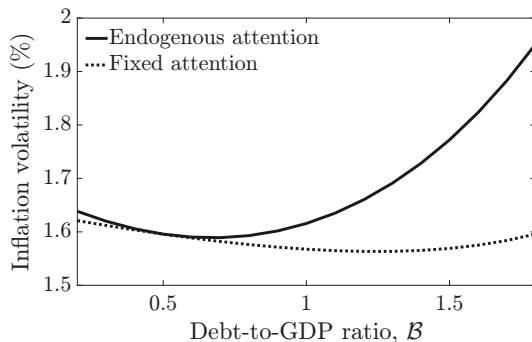
- How important is endogenous attention for this result?

Inflation Volatility with Exogenous Attention Allocation



- Inflation volatility is falling with $\mathcal{B}$ when attention is fixed (fixed at its $\mathcal{B} = 0.5$ value)

Inflation Volatility with Exogenous Attention Allocation



- ▶ Inflation volatility is falling with $\mathcal{B}$ when attention is fixed (fixed at its $\mathcal{B} = 0.5$ value)
- ▶ Why?

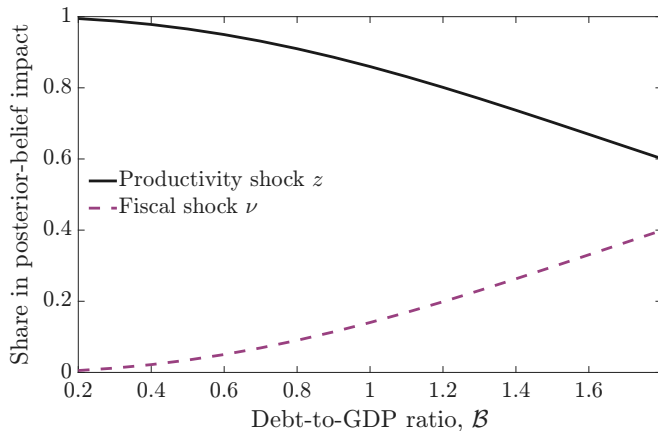
The Role of Endogenous Attention

Firms choose their information structure:

- ▶ how much information to acquire
- ▶ and the optimal signal loads endogenously on productivity- and fiscal-related components of the state

We now decompose firms' posterior beliefs into whether they are mostly driven by productivity or by fiscal shocks

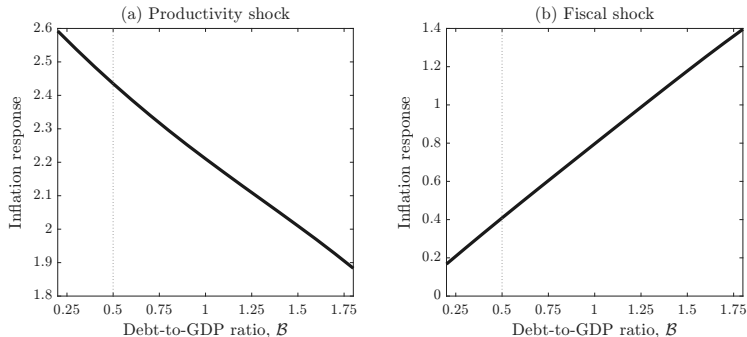
What do Firms Pay Attention to?



- ▶ Posterior beliefs are mostly driven by productivity shocks, especially at low $\mathcal{B}$
- ▶ Fiscal shocks become more important as $\mathcal{B}$ increases

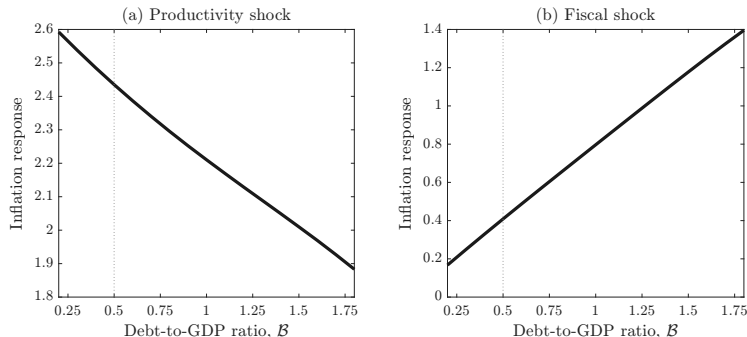
Inflation Response to Shocks

Cumulative inflation response (first four quarters) to the shocks for different $\mathcal{B}$:



Inflation Response to Shocks

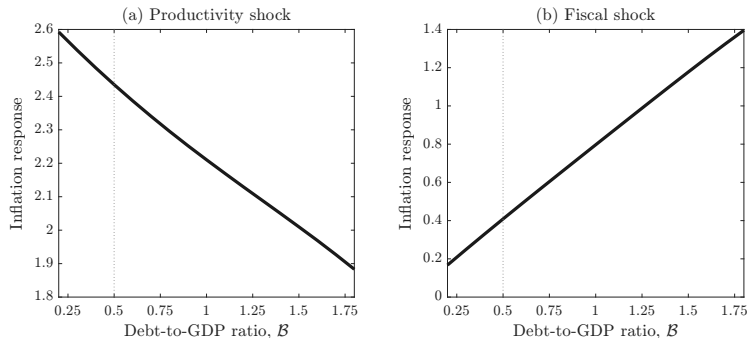
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↪ productivity shocks become less inflationary as $\mathcal{B}$ increases because of endogenous tax adjustment ↪ predicts inflation volatility decrease

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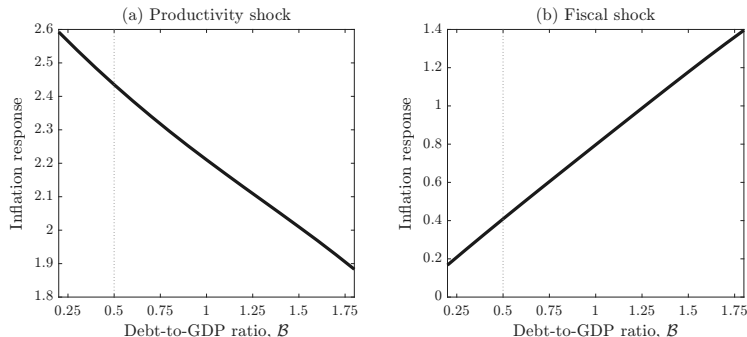
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- fiscal shocks become more inflationary due to $\gamma(\mathcal{B})$

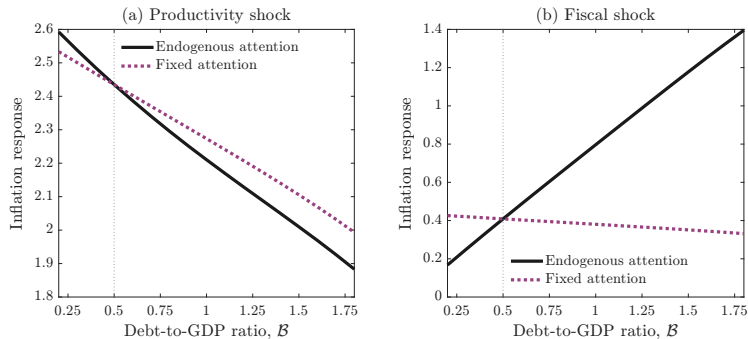
Inflation Response to Shocks

Cumulative inflation response (first four quarters) to the shocks for different $\mathcal{B}$:



- productivity shocks become less inflationary as $\mathcal{B}$ increases because of endogenous tax adjustment → predicts inflation volatility decrease
- fiscal shocks become more inflationary due to $\gamma(\mathcal{B})$
- more attention to fiscal shocks when $\mathcal{B}$ increases → inflation volatility increases

Endogenous Attention Shapes Inflation Response to Shocks



- ↪ if we **fix attention allocation**, firms care mostly about **productivity shocks** which become **less inflationary** as $\mathcal{B}$ increases
- ↪ as firms do not pay attention to fiscal shocks, their effects are practically constant in $\mathcal{B}$

Endogenous attention renders effects of fiscal policy state dependent

1. More aggressive fiscal adjustment lowers inflation volatility at low $\mathcal{B}$, but increases it at high $\mathcal{B}$ [► Details](#)
2. Tax hikes coinciding with high uncertainty are especially inflationary at high $\mathcal{B}$ [► Details](#)

Conclusion

- ▶ We develop a model in which fiscal policy directly affects firms' desired prices through revenue taxes, and firms choose attention endogenously
 - ▶ Fiscal policy matters for firms' attention: it changes both the stakes of mispricing and the sensitivity of desired prices to fiscal conditions
 - ▶ Attention shapes passthrough, inflation dynamics and volatility
- ↪ **Policy implications depend on endogeneity of attention allocation**

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- ↪ **Policy implications depend on endogeneity of attention allocation**

Thank you!

Appendix

Robustness: 41 specifications

	$\hat{\beta}_1 > 0$	Significant at α (one-sided)	
		10%	5%
All 41 specifications	41/41	39/41	37/41
Controls / FEs (25)	25/25	25/25	25/25
Leave-one-country-out (11)	11/11	9/11	9/11
Estimator variations (5)	5/ 5	5/ 5	3/ 5

Distribution of $\hat{\beta}_1$ across specs:

	min	median	mean	max
$\hat{\beta}_1 (\times 10^4)$	+2.7	+6.9	+6.4	+9.6

► Back

Leave-one-country-out

Dropped country	$\hat{\beta}_1$ ($\times 10^4$)	SE ($\times 10^4$)	$p(\hat{\beta}_1 > 0)$	n
None (baseline)	7.19***	2.98	0.008	62,836
GR	9.60***	1.92	0.000	58,211
ES	7.91***	2.95	0.004	53,771
IE	7.59***	2.99	0.006	59,058
AT	7.32***	2.97	0.007	57,689
FI	7.27***	2.98	0.007	59,755
NL	7.45***	3.09	0.008	56,763
PT	7.00**	3.05	0.011	58,163
BE	7.01**	3.07	0.011	58,267
SK	6.87**	3.06	0.012	59,782
DE	5.61*	3.50	0.055	53,550
IT	2.73	2.35	0.123	53,351

***, **, *: $p < 0.01, 0.05, 0.10$ (one-sided). $\hat{\beta}_1 > 0$ in every leave-one-out specification. [► Back](#)

Estimator variations

Estimator	$\hat{\beta}_1$ ($\times 10^4$)	SE ($\times 10^4$)	p
Baseline (cluster SE, weighted)	7.19***	2.98	0.008
HC1 robust SEs (no clustering)	7.19***	3.08	0.010
Unweighted	4.17*	2.93	0.077
Winsorize debt 5/95	7.67***	2.75	0.003
Median regression	8.42***	0.54	0.000
Huber M-estimator	3.17***	0.90	0.000

Median and Huber regressions do not use cluster-robust standard errors; SEs reported here are sandwich-based. [► Back](#)

Robustness: production taxes instead of debt-to-GDP

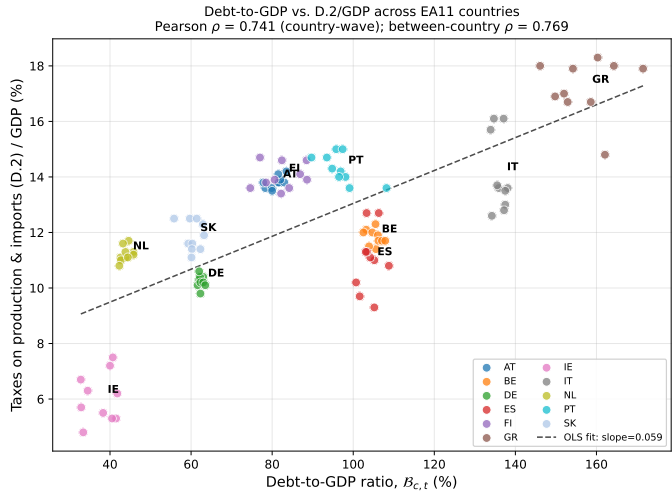
What we use. Eurostat D.2 (*Taxes on production and imports*), as % of GDP, lagged one quarter. Source: Eurostat (gov_10q_ggnfa)

What D.2 measures: Compulsory payments levied on the production and importation of goods and services, **payable irrespective of profits made**. Includes:

- ▶ D.211 – Value-added taxes (VAT)
- ▶ D.212 / D.214 – Import duties, excise duties (alcohol, tobacco, fuel), stamp duties
- ▶ D.29 – Other production taxes (business assets, certain payroll taxes, licenses)

Result. Replacing $\mathcal{B}_{c,t}$ with $D.2_{c,t}/\text{GDP}$ in the baseline:

	Coef.	SE
Δmc^e	0.178***	(0.042)
$\Delta mc^e \times D.2/\text{GDP}$	0.0069**	(0.0036)
Implied passthrough at min D.2/GDP (4.8%)	0.21	
Implied passthrough at max D.2/GDP (18.3%)	0.31	



Decreasing returns to scale

Production $Y_j = Z_j L_j^\alpha$ with $\alpha \in (0, 1]$.

Define $\zeta = (1 - \alpha)/\alpha$ and $a_\alpha = 1/(1 + \eta\zeta) = \alpha/[\alpha + \eta(1 - \alpha)]$.

	$\beta(\mathcal{B})$	Movement in $\mathcal{B}$
Full information	$a_\alpha(1 + \zeta) = \frac{1}{\alpha + \eta(1 - \alpha)}$	constant
Misspec. attribution	$\frac{a_\alpha(1 + \zeta)}{\delta}$	constant
Endogenous attention	$a_\alpha(1 + \zeta) \left[1 + \frac{\gamma(\mathcal{B})^2 \sigma_\tau^2}{(1 + \zeta)^2 \sigma_z^2} \right]$	rising

$\Rightarrow$ DRS decreases β , but leaves dependency on $\mathcal{B}$ unchanged. ► Back

Households and Final Good Producer ▶ back

Representative household solves:

$$\max_{\{C_t, L_t, B_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\log C_t - L_t]$$

$$\text{subject to} \quad P_t C_t + B_t = (1 + i_{t-1}) B_{t-1} + W_t L_t + Profits_t, \quad B_0 \text{ given}$$

▶ $\mathbb{E}_0$ is the full information expectations operator

$\hookrightarrow P_t C_t = W_t$: nominal consumption equals nominal wage, standard Euler equation

Competitive final good producer (CES aggregator):

$$\max_{\{Y_t, (Y_{j,t}^d)_{j \in [0,1]}\}} P_t Y_t - \int_0^1 P_{j,t} Y_{j,t}^d dj \quad \text{s.t.} \quad Y_t = \left(\int_0^1 (Y_{j,t}^d)^{1-\eta} dj \right)^{\frac{1}{1-\eta}}$$

$$\Rightarrow \quad Y_{j,t}^d = \left(\frac{P_{j,t}}{P_t} \right)^{-\eta} Y_t \quad \text{where} \quad P_t = \left[\int_0^1 P_{j,t}^{1-\eta} dj \right]^{\frac{1}{1-\eta}}.$$

With profit and payroll taxes instead of revenue taxes, the firm's problem becomes:

$$\max_{P_{j,t}} \left[(1 - \mathcal{T}_{\pi,t})P_{j,t} - (1 - \mathcal{T}_{\pi,t})(1 + \mathcal{T}_{\ell,t})MC_t \right] \left(\frac{P_{j,t}}{P_t} \right)^{-\eta} Y_t.$$

Proposition

Suppose that for every t ,

$$\mathcal{T}_{\pi,t} = \mathcal{T}_{r,t}, \quad 1 + \mathcal{T}_{\ell,t} = \frac{1}{1 - \mathcal{T}_{r,t}} \quad \Leftrightarrow \quad \mathcal{T}_{\ell,t} = \frac{\mathcal{T}_{r,t}}{1 - \mathcal{T}_{r,t}}. \quad (2)$$

Then:

- (i) firm j 's pricing problem is identical to the pricing problem under the revenue tax*
- (ii) the full-information ideal price and the second-order approximation to profits are unchanged; and*
- (iii) government tax revenues are identical period by period.*

Model solution and calibration [▶ back](#)

- ▶ Solve the model as in Afrouzi/Yang but with public signal [▶ Details](#)
 - ▶ Guess law of motion, solve attention problem, solve GE, iterate until convergence
 - ▶ Public signal: firms observe past prices $\leftrightarrow$ firms track stationary optimal price change, instead of non-stationary optimal price

Parameter	Description	Value
$\mathcal{B}$	Steady-state debt level	varies
ϕ_b	Fiscal feedback on debt in tax rule	2 (and 2.5)
ϕ_π	Taylor-rule response to inflation	1.5
$\bar{i}$	Steady-state nominal interest rate	0.1
η	Elasticity of substitution	6
ω	Marginal information cost	0.0011
ρ_z	Persistence of productivity shock	0.40
ρ_ν	Persistence of tax shock	0.40
σ_z	Std. dev. of productivity shock	0.0375
σ_ν	Std. dev. of tax shock	0.1064

[▶ Details](#)

The Dynamic Rational Inattention Problem ▶ back

$$\begin{aligned}
 & \max_{\{\kappa_{j,t}, S_{j,t}, P_{j,t}(\mathcal{I}_{j,t}^{\text{afternoon}})\}_{t \geq 0}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \frac{1}{P_t C_t} \left\{ (1 - \mathcal{T}_{r,t}) P_{j,t} Y_{j,t} - W_t L_{j,t} - \omega \kappa_{j,t} \right\} \right] \\
 \text{s.t.} \quad & Y_{j,t} = Y_{j,t}^d && \text{(demand)} \\
 & \mathbb{I}(S_{j,t}; \vec{x}_t \mid \mathcal{I}_{j,t-1}) \leq \kappa_{j,t} && \text{(information constraint)} \\
 & \mathcal{I}_{j,t}^{\text{afternoon}} = \sigma(\mathcal{I}_{j,t-1}, S_{j,t}) && \text{(after private signal)} \\
 & \mathcal{I}_{j,t} = \sigma(\mathcal{I}_{j,t}^{\text{afternoon}}, p_t) && \text{(after aggregate price)} \\
 & \mathcal{J}_t = \sigma(\mathcal{J}_{t-1}, p_t) && \text{(public updating).}
 \end{aligned}$$

- ▶ The current private signal is chosen conditional on $\mathcal{I}_{j,t-1}$.
- ▶ The price $P_{j,t}$ is chosen using $\mathcal{I}_{j,t}^{\text{afternoon}}$, before the free observation of p_t .
- ▶ The state in the information constraint is the state $\vec{x}_t$ outlined below.

State-space representation of the target

The stationary target is the ideal price relative to the lagged aggregate price:

$$\tilde{p}_{j,t}^{\diamond} \equiv p_{j,t}^{\diamond} - p_{t-1} = \Phi_z(\mathbf{L})\epsilon_t^z + \Phi_{\nu}(\mathbf{L})\epsilon_t^{\nu}.$$

- ▶ Since price is non-stationary (unit root from the Taylor rule), we work with $\tilde{p}^{\diamond}$, which is stationary and admits a standard Wold representation.

Following Afrouzi/Yang, we approximate the desired price as an MA(L) process driven by the exogenous shocks $\{z_t, \nu_t\}$. Truncating the Wold representation at lag $\mathbf{L}$, define

$$\begin{aligned}\vec{x}_t &\equiv (\epsilon_t^z, \epsilon_t^{\nu}, \epsilon_{t-1}^z, \epsilon_{t-1}^{\nu}, \dots, \epsilon_{t-\mathbf{L}}^z, \epsilon_{t-\mathbf{L}}^{\nu})' \in \mathbb{R}^n, & n &\equiv 2(\mathbf{L} + 1), \\ \vec{x}_t &= \mathbf{A}\vec{x}_{t-1} + \mathbf{Q}\epsilon_t, & \epsilon_t &\equiv (\epsilon_t^z, \epsilon_t^{\nu})' \sim \mathcal{N}(\mathbf{0}, \text{diag}(\sigma_z^2, \sigma_{\nu}^2)), \\ \tilde{p}_{j,t}^{\diamond} &= \mathbf{H}'\vec{x}_t, & \mathbf{H} &\in \mathbb{R}^n.\end{aligned}$$

- ▶ $\mathbf{A}$ is the block-shift matrix, $\mathbf{Q}$ loads the two innovations into the top two entries of $\vec{x}_t$.
- ▶ $\mathbf{H}$ is endogenous and is pinned down by an outer fixed point (details later).

The firm's problem in covariance form

Define the stationary action $a_{j,t} \equiv p_{j,t} - p_{t-1}$. Since $p_{t-1} \in \mathcal{I}_{j,t-1}$,

$$(p_{j,t} - p_{j,t}^\diamond)^2 = (a_{j,t} - \mathbf{H}' \vec{x}_t)^2.$$

The optimal pricing rule is the conditional-mean tracking rule:

$$a_{j,t} = \mathbf{H}' \mathbb{E}[\vec{x}_t \mid \mathcal{I}_{j,t}^{\text{afternoon}}].$$

The flow payoff-stakes matrix is

$$\Omega(\mathcal{B}) \equiv \mathcal{S}(\mathcal{B}) \mathbf{H} \mathbf{H}', \quad \mathcal{S}(\mathcal{B}) = (1 - i\mathcal{B})(\eta - 1).$$

For Gaussian signals,

$$\mathbb{I}(S_{j,t}; \vec{x}_t \mid \mathcal{I}_{j,t-1}) = \frac{1}{2} \log \frac{\det \Sigma_{t|t-1}^i}{\det \Sigma_t^{i,pp}}.$$

The covariance problem and the public update

The firm chooses the post-private, pre-public covariance $\Sigma_t^{i,pp}$:

$$\begin{aligned} \min_{\{\Sigma_t^{i,pp}\}_{t \geq 0}} \quad & \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \left\{ \frac{1}{2} \text{tr} \left(\Omega(\mathcal{B}) \Sigma_t^{i,pp} \right) + \frac{\omega}{2} \log \frac{\det \Sigma_{t|t-1}^i}{\det \Sigma_t^{i,pp}} \right\} \middle| \mathcal{I}_{j,-1} \right] \\ \text{s.t.} \quad & \mathbf{0} \leq \Sigma_t^{i,pp} \leq \Sigma_{t|t-1}^i, \quad \Sigma_{t+1|t}^i = \mathbf{A} \Sigma_t^{i,p} \mathbf{A}' + \mathbf{Q} \text{diag}(\sigma_z^2, \sigma_\nu^2) \mathbf{Q}'. \end{aligned}$$

After the firm has chosen $\Sigma_t^{i,pp}$, the public signal applies the noiseless update

$$\begin{aligned} \Sigma_t^{i,p} &= (I - G_t^i) \Sigma_t^{i,pp} (I - G_t^i)', \\ G_t^i &\equiv \Sigma_t^{i,pp} c_t (c_t' \Sigma_t^{i,pp} c_t)^{-1} c_t', \quad c_t = (K_t^{agg})' \mathbf{H}. \end{aligned}$$

- The individual firm takes c_t as given.

Dynamic information Euler equation

For a symmetric matrix $X = UDU'$ with eigenvalues $d_1 \geq d_2 \geq \dots \geq d_n \geq 0$, define

$$\text{Max}(X, \omega) \equiv U \text{diag}(\max\{d_k, \omega\}_{k=1}^n) U',$$

$$\text{Min}(X, \omega) \equiv U \text{diag}(\min\{d_k, \omega\}_{k=1}^n) U'.$$

The value-weighted uncertainty matrix is

$$X_t \equiv (\Sigma_{t|t-1}^i)^{1/2} \bar{\Omega}_t (\Sigma_{t|t-1}^i)^{1/2}.$$

The information Euler equation is

$$\Sigma_t^{i,pp} = \omega (\Sigma_{t|t-1}^i)^{1/2} \text{Max}(X_t, \omega)^{-1} (\Sigma_{t|t-1}^i)^{1/2}.$$

The implied chosen capacity is

$$\kappa_{j,t} = \frac{1}{2} \sum_{\ell=1}^n \log \max\left(\frac{d_\ell}{\omega}, 1\right).$$

Private posterior gain and active directions

Let $(d_\ell, u_\ell)_{\ell=1,\dots,n}$ denote the eigenpairs of

$$X_t = (\Sigma_{t|t-1}^i)^{1/2} \bar{\Omega}_t (\Sigma_{t|t-1}^i)^{1/2}, \quad y_\ell \equiv (\Sigma_{t|t-1}^i)^{-1/2} u_\ell.$$

The private posterior gain associated with the new signal $S_{j,t}$ is

$$K_t^{priv} = \sum_{d_\ell > \omega} \left(1 - \frac{\omega}{d_\ell}\right) (\Sigma_{t|t-1}^i y_\ell) y_\ell'.$$

Equivalently,

$$K_t^{priv} = I - \Sigma_t^{i,pp} (\Sigma_{t|t-1}^i)^{-1}.$$

- ▶ The factor $1 - \omega/d_\ell$ is the variance-reduction share along direction ℓ .
- ▶ The number of active directions equals the number of eigenvalues of X_t that exceed ω .
- ▶ In our model with one action per period, there is at most one active direction.

Continuation value

Today's choice of $\Sigma_t^{i,pp}$ enters tomorrow's private prior through the public update and the state law of motion:

$$\begin{aligned}\Sigma_t^{i,p} &= \mathcal{P}_{c_t}(\Sigma_t^{i,pp}) = (I - G_t^i)\Sigma_t^{i,pp}(I - G_t^i)', \\ G_t^i &= \Sigma_t^{i,pp}c_t(c_t'\Sigma_t^{i,pp}c_t)^{-1}c_t', \\ \Sigma_{t+1|t}^i &= \mathbf{A}\mathcal{P}_{c_t}(\Sigma_t^{i,pp})\mathbf{A}' + \mathbf{Q}\text{diag}(\sigma_z^2, \sigma_\nu^2)\mathbf{Q}'.\end{aligned}$$

$\bar{\Omega}_t$ solves:

$$\begin{aligned}\bar{\Omega}_t &= \Omega(\mathcal{B}) \\ &+ \beta(I - G_t^i)'\mathbf{A}'(\Sigma_{t+1|t}^i)^{-1/2}\text{Min}\left((\Sigma_{t+1|t}^i)^{1/2}\bar{\Omega}_{t+1}(\Sigma_{t+1|t}^i)^{1/2}, \omega\right)(\Sigma_{t+1|t}^i)^{-1/2}\mathbf{A}(I - G_t^i).\end{aligned}$$

Aggregate prices as a public signal

The cross-sectional average of firms' afternoon posterior means is

$$\bar{x}_t = m_{t|t-1}^c + K_t^{agg}(\vec{x}_t - m_{t|t-1}^c).$$

The aggregate gain combines information from private histories and the new private signal:

$$\begin{aligned} K_t^{hist} &\equiv (\Sigma_{t|t-1}^c - \Sigma_{t|t-1}^i)(\Sigma_{t|t-1}^c)^{-1}, \\ K_t^{agg} &= K_t^{priv} + (I - K_t^{priv})K_t^{hist}. \end{aligned}$$

Aggregate inflation satisfies

$$\begin{aligned} \pi_t &= \mathbf{H}' m_{t|t-1}^c + \mathbf{H}' K_t^{agg}(\vec{x}_t - m_{t|t-1}^c), \\ \pi_t - \mathbb{E}[\pi_t | \mathcal{J}_{t-1}] &= c_t'(\vec{x}_t - m_{t|t-1}^c), \quad c_t \equiv (K_t^{agg})'\mathbf{H}. \end{aligned}$$

Thus observing π_t is informationally equivalent to observing the scalar noiseless public signal $s_t^\pi = c_t'(\vec{x}_t - m_{t|t-1}^c)$.

Public and private covariance updates

The public covariance update is

$$\begin{aligned}\Sigma_t^{c,p} &= (I - G_t^c) \Sigma_{t|t-1}^c (I - G_t^c)', \\ G_t^c &\equiv \Sigma_{t|t-1}^c c_t (c_t' \Sigma_{t|t-1}^c c_t)^{-1} c_t' .\end{aligned}$$

The private covariance update starts from the afternoon covariance $\Sigma_t^{i,pp}$:

$$\begin{aligned}\Sigma_t^{i,p} &= (I - G_t^i) \Sigma_t^{i,pp} (I - G_t^i)', \\ G_t^i &\equiv \Sigma_t^{i,pp} c_t (c_t' \Sigma_t^{i,pp} c_t)^{-1} c_t' .\end{aligned}$$

The post-evening covariances propagate forward as

$$\begin{aligned}\Sigma_{t+1|t}^c &= \mathbf{A} \Sigma_t^{c,p} \mathbf{A}' + \mathbf{Q} \text{diag}(\sigma_z^2, \sigma_\nu^2) \mathbf{Q}', \\ \Sigma_{t+1|t}^i &= \mathbf{A} \Sigma_t^{i,p} \mathbf{A}' + \mathbf{Q} \text{diag}(\sigma_z^2, \sigma_\nu^2) \mathbf{Q}' .\end{aligned}$$

Steady-state equilibrium with public price history

A stationary rational-inattention equilibrium with public price history is a tuple

$$(\mathbf{H}, \Sigma^i, \Sigma^c, \Sigma^{i,pp}, \bar{\Omega}, K^{priv}, K^{hist}, K^{agg}, G^c, G^i, c)$$

such that

$$\mathbf{0} \leq \Sigma^{i,pp} \leq \Sigma^i \leq \Sigma^c, \quad c = (K^{agg})' \mathbf{H},$$

and the following conditions hold:

1. The information Euler equation holds:

$$\Sigma^{i,pp} = \omega(\Sigma^i)^{1/2} \text{Max} \left((\Sigma^i)^{1/2} \bar{\Omega} (\Sigma^i)^{1/2}, \omega \right)^{-1} (\Sigma^i)^{1/2}.$$

2. The continuation-value recursion holds with G^i .
3. The public and private covariance fixed points hold with G^c and G^i .
4. The target loading $\mathbf{H}$ is consistent with the implied aggregate equilibrium.

Steady-state continuation value and covariance fixed point

The steady-state continuation value is

$$\begin{aligned}\bar{\Omega} &= \mathcal{S}(\mathcal{B})\mathbf{H}\mathbf{H}' + \beta(I - G^i)' \mathbf{A}' (\Sigma^i)^{-1/2} \\ &\quad \times \text{Min}\left((\Sigma^i)^{1/2} \bar{\Omega} (\Sigma^i)^{1/2}, \omega\right) (\Sigma^i)^{-1/2} \mathbf{A} (I - G^i).\end{aligned}$$

The covariance fixed point is

$$\begin{aligned}\Sigma^c &= \mathbf{A}(I - G^c) \Sigma^c (I - G^c)' \mathbf{A}' + \mathbf{Q} \text{diag}(\sigma_z^2, \sigma\nu^2) \mathbf{Q}', \\ \Sigma^i &= \mathbf{A}(I - G^i) \Sigma^{i,pp} (I - G^i)' \mathbf{A}' + \mathbf{Q} \text{diag}(\sigma_z^2, \sigma\nu^2) \mathbf{Q}'.\end{aligned}$$

Here

$$K^{agg} = K^{priv} + (I - K^{priv})K^{hist}, \quad K^{hist} = (\Sigma^c - \Sigma^i)(\Sigma^c)^{-1}, \quad c = (K^{agg})' \mathbf{H}.$$

Outer fixed point on the target loading

The aggregate state is

$$s_t \equiv (\vec{x}_t, m_{t|t-1}^c, b_{t-1}^R, \pi_{t-1})' \in \mathbb{R}^{2n+2}, \quad b_t^R \equiv b_t - p_t.$$

Aggregate inflation is

$$\pi_t = \mathbf{P} s_t, \quad \mathbf{P} = (\mathbf{H}' K^{agg}, \mathbf{H}' (I - K^{agg}), 0, 0).$$

The consumption row vector $\mathbf{C}$ is pinned down by

$$\mathbf{C}(I_{2n+2} - \mathbf{F}) = \mathbf{P}(\mathbf{F} - \phi_\pi I_{2n+2}), \quad s_{t+1} = \mathbf{F} s_t + \mathbf{G} \epsilon_{t+1}.$$

Given the solved system $(\mathbf{F}, \mathbf{G}, \mathbf{P}, \mathbf{C})$, simulate basis shocks to recover

$$\tilde{p}_{j,t}^{\diamond, GE} = \pi_t + c_t - z_t + \gamma(\mathcal{B}) \phi_b b_{t-1}^R + \gamma(\mathcal{B}) \nu_t = (\mathbf{H}^{new})' \vec{x}_t.$$

The outer fixed point requires

$$\mathbf{H}^{new} = \mathbf{H}.$$

Numerical algorithm

The equilibrium is solved by nested iteration.

1. Guess $\mathbf{H}^{(0)} \in \mathbb{R}^n$ and $\mathbf{C}^{(0)} \in \mathbb{R}^{2n+2}$.

2. Given $\mathbf{H}^{(r)}$, form

$$\Omega^{(r)} = \mathcal{S}(\mathcal{B})\mathbf{H}^{(r)}\mathbf{H}^{(r)'}$$

3. Iterate the joint covariance recursion: compute $\Sigma^{i,pp}$, K^{priv} , K^{hist} , K^{agg} , c , G^c , G^i , Σ^c , Σ^i , and $\bar{\Omega}$ until convergence.

4. Given (K^{agg}, G^c) , form $\mathbf{P}$ and $\mathbf{F}$ and iterate on

$$\mathbf{C}(I_{2n+2} - \mathbf{F}) = \mathbf{P}(\mathbf{F} - \phi_\pi I_{2n+2})$$

until convergence.

5. Compute $\mathbf{H}^{new}$ by impulse simulation of $s_{t+1} = \mathbf{F}s_t + \mathbf{G}\epsilon_{t+1}$.

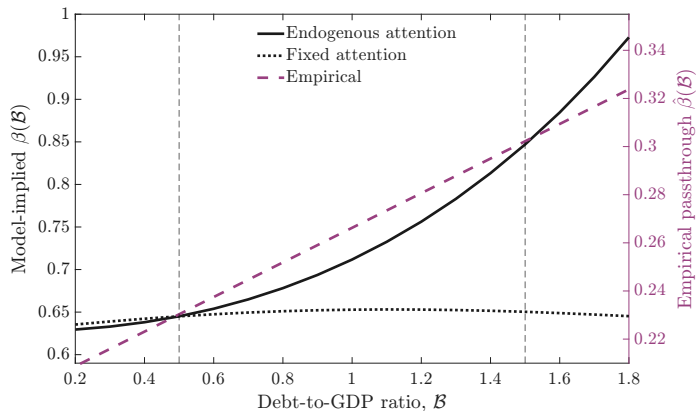
6. Update

$$\mathbf{H}^{(r+1)} = (1 - \theta)\mathbf{H}^{(r)} + \theta\mathbf{H}^{new}, \quad \theta \in (0, 1],$$

until $\|\mathbf{H}^{(r+1)} - \mathbf{H}^{(r)}\| < \varepsilon_{outer}$.

Calibration: Details

- ▶ Target the slope of $\beta(\mathcal{B})$ from $\mathcal{B} = 0.5$ to 1.5



- ▶ Basically flat when fixing attention ▶ [back](#)

Fiscal adjustment and inflation volatility

- ▶ A higher fiscal response, ϕ_b , affects firms through **two opposing channels**:
 1. **Debt-stabilization channel**: stronger tax adjustment makes debt less persistent
 - ↪ firms face less uncertainty about desired prices

Fiscal adjustment and inflation volatility

- ▶ A higher fiscal response, ϕ_b , affects firms through **two opposing channels**:
 1. **Debt-stabilization channel**: stronger tax adjustment makes debt less persistent
 - ↪ firms face less uncertainty about desired prices
 2. **Tax-sensitivity channel**: stronger tax adjustment makes desired prices more sensitive to debt and tax shocks
 - ▶ Desired price moves with the tax wedge:
$$p_{j,t}^{\diamond} = mc_t + \gamma(\mathcal{B}) \tau_{r,t}, \quad \tau_{r,t} = \phi_b b_{t-1}^R + \nu_t$$
 - ↪ a higher ϕ_b raises the payoff relevance of fiscal information

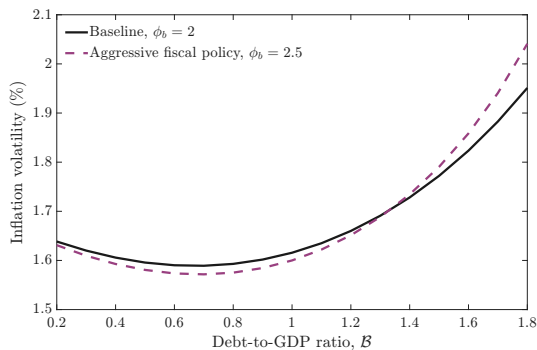
Low debt vs. high debt: why the sign flips

- ▶ Low $\mathcal{B}$: tax wedge is small, fiscal shocks are not very important for pricing
 - ↪ the **stabilization effect dominates**: stronger ϕ_b reduces uncertainty about debt
 - ↪ firms pay **less attention** to fiscal conditions and adjust prices less aggressively
 - ↪ **inflation volatility falls**

Low debt vs. high debt: why the sign flips

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 - ↪ the stabilization effect dominates: stronger ϕ_b reduces uncertainty about debt
 - ↪ firms pay less attention to fiscal conditions and adjust prices less aggressively
 - ↪ inflation volatility falls
- ▶ High $\mathcal{B}$: tax wedge is large, fiscal shocks are more relevant for desired prices
 - ↪ tax-sensitivity effect dominates: a higher ϕ_b makes taxes matter more for pricing
 - ↪ firms pay more attention to fiscal conditions and reprice more aggressively
 - ↪ inflation volatility rises

Complementarity between Fiscal Policy and Endogenous Attention



- Higher ϕ_b is stabilizing at low debt but raises inflation volatility at high debt.

↪ complementarity between fiscal policy and attention

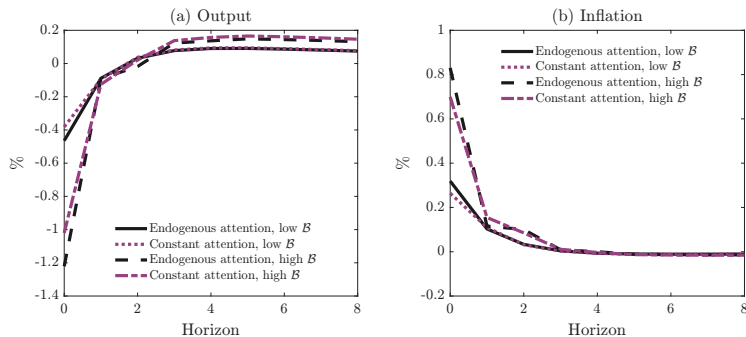
► back

The Role of Uncertainty

Assume that at the time a fiscal shock occurs, prior uncertainty also **increases**

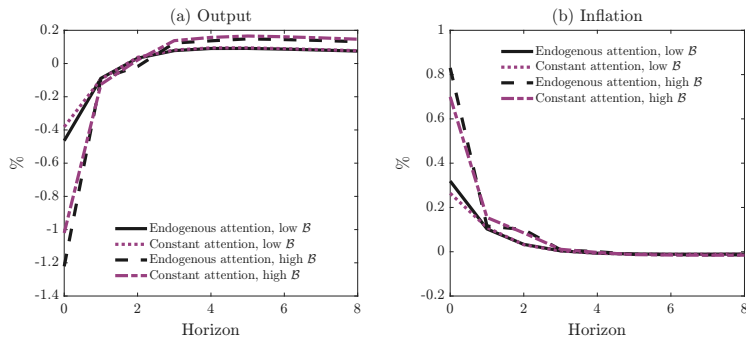
The Role of Uncertainty

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The Role of Uncertainty

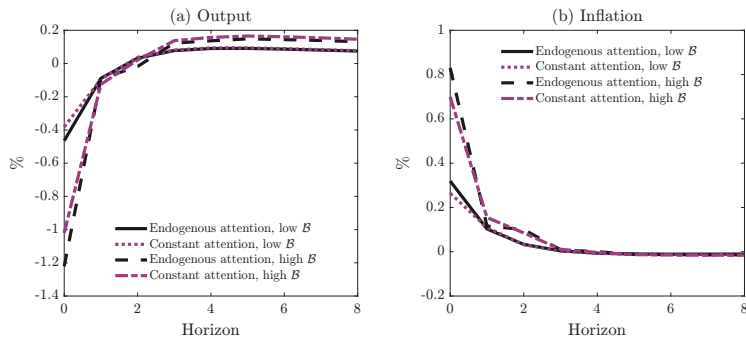
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Assume that at the time a fiscal shock occurs, prior uncertainty also increases



- uncertainty matters more at high $\mathcal{B}$ ($\Delta\pi_0^{\mathcal{B}_{high}} \approx 2 \cdot \Delta\pi_0^{\mathcal{B}_{low}}$)
- whether taxes are hiked in times of higher uncertainty or not is not that important at low debt levels, but it matters at high debt levels